

Accelerated Reinforcement Learning With Verifiable Excitation for Cubic Convergence

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Abstract—This article proposes and analyzes an accelerated reinforcement learning (RL) algorithm for discrete-time linear systems with unknown dynamics. The method achieves cubic convergence, improving upon the quadratic rates of existing policy iteration (PI)-based RL algorithms, and it does not rely on persistency of excitation (PE). The value function matrix is computed through a midpoint-centered Lyapunov equation, which provides a third-order Newton-type update and ensures fast convergence. To enable data-driven implementation without PE, we introduce a verifiable excitation condition based on filter outputs. This condition is easily testable online and guarantees that the collected data are sufficiently informative for solving the required least-squares equations. Within this framework, we further propose a data-driven procedure to compute an initial admissible control gain, where a bisection rule is used to determine the maximal feasible discount factor. Theoretical analysis establishes cubic convergence of the accelerated algorithm and proves the effectiveness of the designed excitation condition. Finally, we verify the proposed method through Monte Carlo experiments and comparative simulations with existing methods.

Index Terms—Cubic convergence, data-driven control, optimal control, reinforcement learning (RL), verifiable excitation.

I. INTRODUCTION

REINFORCEMENT learning (RL) has been extensively studied as a framework that enables autonomous agents to learn optimal policies that minimize long-term costs in unknown environments [1], [2], [3], [4]. In contrast to traditional iteration/control, a remarkable feature of RL is its ability to optimize performance directly from data without requiring explicit knowledge of the system dynamics. This data-driven capability provides a promising foundation for model-free optimal control and offers a bridge between data-driven control and classical optimal control.

Different contributions have analyzed RL for various types of dynamic systems, e.g., networked control systems [5], [6], nonlinear systems [7], game dynamics systems [8], stochastic systems [9], [10], and multiagent systems [11], [12]. Among

these works, Jiang et al. [5], [6], Shen et al. [7], Wu et al. [8], Xue et al. [10], and Donge et al. [11] proposed policy iteration (PI) algorithms, while Jiang et al. [9] and Li et al. [12] focused on value iteration (VI) algorithms. These algorithms differ in two aspects. First, VI performs single-step updates to the value function based on the Bellman optimality principle, with the control policy implicitly derived, while PI alternates between policy evaluation and improvement for explicit policy updates [13]. Second, VI exhibits linear convergence without requiring an admissible initial policy [14], whereas PI achieves quadratic convergence given a stabilizing initial policy. These differences have motivated recent studies on the tradeoff between PI and VI, such as hybrid PI [15], [16], λ -PI [17], bias-PI [18], scaling PI [19], and homotopic PI [20], [21]. Although these methods are promising and address different problems, they share certain common characteristics in both algorithmic design and performance, which can be analyzed from the following perspectives.

- 1) *Convergence Rate*: The convergence rate is critical in practice. For example, in autonomous vehicle systems, the automated system's decisions can be generated using RL, as discussed in [22]. In such scenarios, algorithms with slow convergence, such as VI, may struggle to adapt to dynamic environments in time. In contrast, PI-based methods, benefiting from quadratic convergence owing to Newton-type iterations, can quickly produce near-optimal control policies with only a few iterations. Besides these iterative schemes, another line of work accelerates learning by modifying the parameter update laws, including gradient-based updates [23] and recursive least-squares type schemes [24]. For example, Zhao et al. [25] developed an asynchronous parallel zero-order policy gradient method that achieves computational acceleration with rigorous theoretical guarantees. Xu et al. [26] proposed a recursive least-squares algorithm in which the parameters are updated upon each new data point, thus accelerating the convergence process. These methods mainly exploit first-order information, and their practical convergence speed is affected by the choice of step sizes or forgetting factors. As a result, they typically do not attain the rapid local convergence behavior of high-order Newton-type methods. Several higher-order variants of the Newton method [27], [28], such as Halley's method [29], have been proposed to further improve convergence speed. However, applying these methods to accelerate RL has received limited attention, primarily due to their reliance

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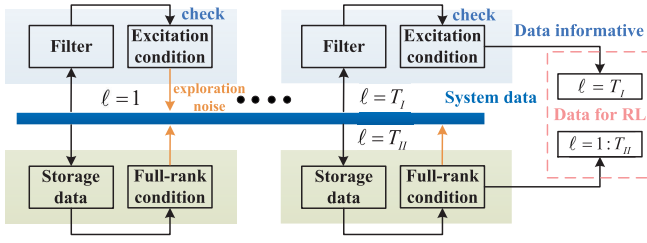


Fig. 1. Scheme illustrating the basic idea of our data-driven RL approach. The top blue part shows our method, which uses data at time $\ell = T_I$ once the excitation condition is satisfied. The bottom green part shows existing methods, which require data from all time steps $\ell = 1 : T_{II}$.

on higher-order derivatives, which are difficult to obtain in data-driven settings. Inspired by this, we aim to develop an accelerated RL algorithm to achieve fast convergence.

- 2) *Data Informativity*: Most existing RL methods [5], [6], [7], [8], [9], [10], [11], [12], [13], [14] rely on persistence of excitation (PE) for data-driven control design, where the excitation condition is often ensured heuristically. Although both the PE condition and its associated rank conditions are often difficult to check in online implementations, they are, in principle, verifiable since they can be assessed once a sufficient data segment is available. However, such verifications typically require constructing large data matrices from past trajectories, for example, the rank condition in [13, Lemma 6], which is computationally expensive. As highlighted in a recent survey [30], there does not exist a systematic way to ensure the PE condition in RL. Several efforts have been made to relax this requirement, including concurrent learning [31], finitely excited learning [32], and memory-efficient learning [33], [34]. However, these approaches have been developed for continuous-time systems, and their direct application to discrete-time settings remains limited. To address this gap, as shown in Fig. 1, our objective is to relax the PE and rank conditions used in existing data-storage-based RL algorithms. Instead of requiring PE over all time or a full-rank condition on a large data matrix constructed from stored trajectories, we design an initially exciting condition and derive an equivalent excitation test in terms of the filter outputs, which can be monitored online without storing historical data.
- 3) *Initialization Step*: Constructing an admissible initial policy is essential, as PI-based RL algorithms require a stabilizing starting point to ensure that policy evaluation is well-posed. Existing RL methods for obtaining such an initial policy generally fall into two categories. The first employs VI to generate an admissible policy before applying PI. For example, Gao et al. [16] proposed a hybrid PI algorithm that addresses denial-of-service attacks and removes the requirement for an admissible control gain. Wang et al. [35] developed a parallel hybrid PI algorithm that efficiently solves the mode-coupled Riccati equations and accelerates the iterative process. The second embeds tuning parameters into the

PI framework to guide the construction of admissible control policies, with representative variants, including λ -PI [17], bias-PI [18], scaling PI [19], and homotopic PI [20], [21]. Although these parameterized schemes are effective, their performance is sensitive to the choice of tuning parameters. For example, scaling PI [19] and homotopic PI [20], [21] require two tuning parameters, whereas discount-PI [36], [37] uses one. Moreover, as noted in [21], discount-PI lacks an explicit update rule for the discount factor, and its effective step size is not formally characterized, which complicates implementation. To reduce the dependence on manually selected parameters, we develop a bisection-based data-driven procedure that computes an admissible initial control policy under the proposed initially exciting framework, enabling online implementation without storing historical data.

Motivated by the above discussions, our objective is to advance existing data-driven RL algorithms for discrete-time linear systems by addressing the aforementioned issues. Our specific contributions are as follows.

- 1) We develop a novel accelerated RL algorithm that operates in both model-based and data-driven scenarios. We prove that the algorithm exhibits cubic convergence, which distinguishes it from existing PI [5], [6], [8], [11], [38] with quadratic convergence and VI [9], [12] with linear convergence.
- 2) We design a data-driven algorithm supported by an excitation condition that is easily verifiable online and does not require storing historical data. We establish a necessary and sufficient condition to determine whether the system data is sufficiently informative for data-driven control design. Unlike existing methods [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], our formulation does not rely on the PE condition.
- 3) We propose a data-driven method for computing admissible control policies, where the update rule for the discount factor is explicitly specified using a bisection algorithm.

The remaining part of this article is organized as follows. Section II presents the preliminaries and problem formulation. In Section III, we propose a model-based algorithm and analyze its convergence. Section IV develops a data-driven algorithm and introduces an online verifiable excitation condition. Section V presents a data-driven method for computing an admissible control gain. Simulation results are provided in Section VI. Finally, Section VII concludes the article.

Notations: Throughout this article, $\mathcal{R}^{n \times n}$ and $\mathcal{R}^n$ denote the set of $n \times n$ real matrices and the n -dimensional Euclidean space, respectively. $\lambda(\bullet)$ denotes the spectral radius, and I_n is the $n \times n$ identity matrix. $\kappa_i(\bullet)$ denotes the i th eigenvalue of the matrix. $\otimes$ denotes the Kronecker product. $\|\bullet\|$ is the Euclidean norm. For a vector $\chi \in \mathcal{R}^n$, $\text{vecv}(\chi) = [\chi_1^2, \chi_1\chi_2, \dots, \chi_1\chi_n, \chi_2^2, \dots, \chi_{n-1}\chi_n, \chi_n^2]^T \in \mathcal{R}^{(n(n+1))/2}$. For a symmetric matrix $\mathcal{Q} = [q_{ij}] \in \mathcal{R}^{n \times n}$, $\text{vecs}(\mathcal{Q}) = [q_{11}, 2q_{12}, \dots, 2q_{1n}, q_{22}, \dots, 2q_{n-1,n}, q_{n,n}]^T \in \mathcal{R}^{(n(n+1))/2}$ and $\text{vec}(\mathcal{Q}) = [\mathcal{Q}_1, \mathcal{Q}_2, \dots, \mathcal{Q}_n] \in \mathcal{R}^n$, where $\mathcal{Q}_i$ denotes i th

column of the matrix $\mathcal{Q}$. The duplication matrix $D_n \in \mathcal{R}^{n^2 \times (n(n+1))/2}$ satisfies $\text{vec}(\mathcal{Q}) = D_n \text{vecs}(\mathcal{Q})$.

II. PRELIMINARIES AND PROBLEM STATEMENT

Consider a discrete-time linear system

$$x(\ell + 1) = Ax(\ell) + Bu(\ell) \quad (1)$$

where $x(\ell) \in \mathcal{R}^n$ and $u(\ell) \in \mathcal{R}^m$ denote the system state and control input, respectively. $A \in \mathcal{R}^{n \times n}$ and $B \in \mathcal{R}^{n \times m}$ are unknown system matrices. Assume that (A, B) is controllable. The control policy $u(\ell) \triangleq -Kx(\ell)$ is designed to minimize the following cost function:

$$V(x(\ell)) = \sum_{\tau=\ell}^{\infty} (x^T(\tau) \mathcal{Q}x(\tau) + u^T(\tau) Ru(\tau)) \quad (2)$$

where weight matrices $\mathcal{Q} = \mathcal{Q}^T \in \mathcal{R}^{n \times n} \geq 0$ and $R = R^T \in \mathcal{R}^{m \times m} > 0$. The optimal control policy, denoted by $u^*(\ell) \triangleq \arg_{u(\ell)} \min V(x(0))$, based on optimal control theory [39], is given by

$$u^*(\ell) = -(R + B^T P^* B)^{-1} B^T P^* A x(\ell) = -K^* x(\ell) \quad (3)$$

where $K^* \in \mathcal{R}^{m \times n}$ denotes the optimal feedback gain and value function matrix $P^* = (P^*)^T \in \mathcal{R}^{n \times n} > 0$ is the unique stabilizing solution of the following algebraic Riccati equation (ARE):

$$A^T P^* A - P^* - A^T P^* B (R + B^T P^* B)^{-1} B^T P^* A = -\mathcal{Q}. \quad (4)$$

We now provide the necessary definitions and lemmas.

Definition 1 (Admissible Control Policy): A state feedback policy $u(\ell) = -Kx(\ell)$ is said to be admissible for system (1) with cost function (2) if $A - BK$ is Schur stable and the cost function (2) remains finite.

Lemma 1: [40] Let $K^{[0]}$ be any admissible control gain, and let $P^{[k]}$ be the solution of the Lyapunov equation

$$P^{[k]} = (K^{[k]})^T R K^{[k]} + \mathcal{Q} + (A - BK^{[k]})^T P^{[k]} (A - BK^{[k]}) \quad (5)$$

and $K^{[k+1]} = (R + B^T P^{[k]} B)^{-1} B^T P^{[k]} A$. Then, the following properties hold.

- 1) $A - BK^{[k]}$ is Schur stable.
- 2) $P^* \leq P^{[k+1]} \leq P^{[k]}$.
- 3) $\lim_{k \rightarrow \infty} K^{[k]} = K^*$ and $\lim_{k \rightarrow \infty} P^{[k]} = P^*$.

Lemma 2: [40, Theorem 2]

The algorithm presented in Lemma 1 exhibits quadratic convergence, i.e.,

$$\|P^{[k+1]} - P^*\| \leq c \|P^{[k]} - P^*\|^2 \quad (6)$$

for some scalar $c > 0$.

Note that quadratic convergence of the algorithm in Lemma 1 is due to its update corresponding to a Newton step applied to ARE (4). However, it requires an initial admissible control gain. In contrast, VI algorithm [41], which updates $P^{[k+1]}$ via $P^{[k+1]} = (K^{[k]})^T R K^{[k]} + \mathcal{Q} + (A - BK^{[k]})^T P^{[k]} (A - BK^{[k]})$ achieves only first-order convergence. This is because it replaces the Newton correction with a single forward propagation step. While it can be initialized from an arbitrary $K^{[0]}$, this

generality comes at the cost of slower convergence. Motivated by this, we formulate the following problem.

Problem: Design both model-based and data-driven accelerated RL algorithms to obtain the optimal control policy (3), such that: 1) cubic convergence is achieved; 2) the PE condition is relaxed in the data-driven setting; and 3) an initial admissible policy is identifiable.

III. MODEL-BASED ALGORITHM DESIGN AND ANALYSIS

This section proposes a model-based accelerated RL algorithm to find K^* and address the *Problem (1)*. The accelerated algorithm: 1) computes a provisional value function matrix, 2) forms the midpoint and updates the control gain, and 3) refines the value function matrix by solving a Lyapunov equation centered at the midpoint. The cubic convergence of the algorithm is guaranteed.

To design a model-based accelerated RL algorithm, we consider a Banach space $\mathbb{P} \subset \{P > 0 \in \mathcal{R}^{n \times n} | P = P^T\}$. Define a mapping $\mathcal{G}(P) : \mathbb{P} \rightarrow \mathbb{P}$ as follows:

$$\mathcal{G}(P) \triangleq A^T P A - P - A^T P B (R + B^T P B)^{-1} B^T P A + \mathcal{Q} \quad (7)$$

and let $\mathcal{K}(P) \triangleq (R + B^T P B)^{-1} B^T P A$. The Fréchet derivative of $\mathcal{G}(P)$ at point P is given by

$$\mathcal{G}'_P(X) \triangleq \lim_{\theta \rightarrow 0} \frac{\mathcal{G}(P + \theta X) - \mathcal{G}(P)}{\theta} = \left. \frac{d\mathcal{G}(P + \theta X)}{d\theta} \right|_{\theta=0}.$$

The proposed accelerated algorithm has three steps.

Step 1: Newton Prediction: At iteration k , we first compute a provisional value function matrix $\tilde{P}^{[k+1]}$. By employing the Fréchet derivative of $\mathcal{G}(P^{[k]})$, we obtain

$$\mathcal{G}'_{P^{[k]}}(\tilde{P}^{[k+1]} - P^{[k]}) = -\mathcal{G}(P^{[k]}). \quad (8)$$

To evaluate the derivative $\mathcal{G}'_{P^{[k]}}$ explicitly, define $\mathcal{D}(\theta) \triangleq B^T P A + \theta B^T X A$ and $\mathcal{R}(\theta) \triangleq R + B^T P B + \theta B^T X B$. We have

$$\begin{aligned} \mathcal{G}'_P(X) &= -X + A^T X A - \left. \frac{d\mathcal{D}^T(\theta)}{d\theta} \right|_{\theta=0} \mathcal{R}^{-1}(0) \mathcal{D}(0) \\ &\quad - \mathcal{D}^T(0) \left. \frac{d\mathcal{R}^{-1}(\theta)}{d\theta} \right|_{\theta=0} \mathcal{D}(0) \\ &\quad - \mathcal{D}^T(0) \mathcal{R}^{-1}(0) \left. \frac{d\mathcal{D}(\theta)}{d\theta} \right|_{\theta=0} \\ &= -X + (A - BK)^T X (A - BK). \end{aligned} \quad (9)$$

Substituting (9) into (8), we rewrite the equation as a discrete-time Lyapunov equation

$$\begin{aligned} \tilde{P}^{[k+1]} &= (A - BK^{[k]})^T \tilde{P}^{[k+1]} (A - BK^{[k]}) \\ &\quad + (K^{[k]})^T R K^{[k]} + \mathcal{Q}. \end{aligned} \quad (10)$$

This step solves for $\tilde{P}^{[k+1]}$ via the Lyapunov equation (10), which is structurally identical to (5) in Lemma 1.

Step 2: Midpoint Gain Update: Given the predicted value function matrix $\tilde{P}^{[k+1]}$, define the midpoint $M^{[k]} \triangleq (\tilde{P}^{[k+1]} + P^{[k]})/2$, and update the midpoint gain using

$$\mathcal{K}(M^{[k]}) = (R + B^T M^{[k]} B)^{-1} B^T M^{[k]} A. \quad (11)$$

Step 3: Newton Correction: This step updates the value function matrix $P^{[k+1]}$ and control gain $K^{[k+1]}$. By combining (10) and (11), we obtain

$$\mathcal{G}'_{M^{[k]}}(P^{[k+1]} - P^{[k]}) = -\mathcal{G}(P^{[k]}). \quad (12)$$

This expression yields the following Lyapunov equation:

$$\begin{aligned} P^{[k+1]} - (K^{[k]})^T R K^{[k]} - Q \\ = (A - BK(M^{[k]}))^T P^{[k+1]} (A - BK(M^{[k]})) \\ + (A - BK^{[k]})^T P^{[k]} (A - BK^{[k]}) \\ - (A - BK(M^{[k]}))^T P^{[k]} (A - BK(M^{[k]})). \end{aligned} \quad (13)$$

Then, the control gain $K^{[k+1]}$ is updated as

$$K^{[k+1]} = (R + B^T P^{[k+1]} B)^{-1} B^T P^{[k+1]} A. \quad (14)$$

Algorithm 1 Model-Based Accelerated RL Algorithm

Input: Stabilizing $K^{[0]}$ and a threshold $\epsilon > 0$.

Output: Optimal control gain K^* .

- 1 **Solve** $\tilde{P}^{[k+1]}$ **from** (10).
 - 2 **Solve** $K(M^{[k]})$ **from** (11).
 - 3 **Update** $P^{[k+1]}$ **and** $K^{[k+1]}$ **from** (13) **and** (14).
 - 4 **Stop** **if** $\|P^{[k+1]} - P^{[k]}\| < \epsilon$, **or** set $k \leftarrow k + 1$ **and** go to Step 1.
-

The feasibility of the Lyapunov equations (10) and (13) is analyzed in [28, Proposition 1]. The proposed model-based accelerated RL algorithm is outlined in Algorithm 1.

Remark 1: Algorithm 1 differs from existing model-based RL methods (e.g., those based on Lemma 1) in two key aspects.

1. It introduces a provisional value function matrix $\tilde{P}^{[k+1]}$, obtained from the Lyapunov equation (10) as a Newton prediction of the optimal value matrix. Together with the correction step (13), this design improves the convergence rate, as demonstrated in Theorem 1.

2. The update of $P^{[k+1]}$ is carried out through a Lyapunov equation (13) centered at the midpoint $M^{[k]}$, which is structurally different from the classical policy evaluation (5).

Remark 2: Algorithm 1 solves the ARE (4) using the midpoint Newton method. For the general equation $\mathcal{F}(\nu) = 0$, Step 1 computes the Newton prediction $\nu_N^{[k+1]} = \nu^{[k]} - \mathcal{F}'(\nu^{[k]})^{-1} \mathcal{F}(\nu^{[k]})$. Step 2 takes the midpoint $\nu_M^{[k]} = 0.5(\nu^{[k]} + \nu_N^{[k+1]})$. Step 3 updates at the midpoint as $\nu^{[k+1]} = \nu^{[k]} - \mathcal{F}'(\nu_M^{[k]})^{-1} \mathcal{F}(\nu^{[k]})$. Fig. 2 illustrates these steps for $\mathcal{F}(\nu) = \nu^3$. At iteration k , the midpoint update $\nu^{[k+1]} = 0.52$ is closer to the root than the Newton step $\nu_N^{[k+1]} = 2/3$.

We now prove that Algorithm 1 exhibits cubic convergence.

Theorem 1: Consider the model-based accelerated RL algorithm initialized with an admissible control gain. There exists a neighborhood of P^* such that the algorithm exhibits cubic convergence, i.e.,

$$\|P^{[k+1]} - P^*\| \leq \mathcal{C} \|P^{[k]} - P^*\|^3 \quad (15)$$

for some scalar $\mathcal{C} > 0$.

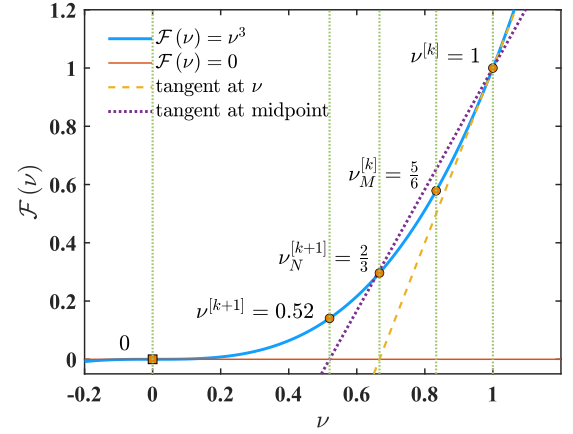


Fig. 2. Illustration of Steps 1–3 for $\mathcal{F}(\nu) = \nu^3$. The yellow line (Step 1) gives the Newton prediction $\nu_N^{[k+1]} = 2/3$. The purple line (Step 3) uses the Jacobian evaluated at the midpoint $\nu_M^{[k]}$ and produces $\nu^{[k+1]} = 0.52$.

Proof: We begin by computing the second-order Fréchet derivative of the mapping $\mathcal{G}(P)$. From (9), it follows that:

$$\mathcal{G}''_P(X, Y) \triangleq \mathcal{T}_Y(\mathcal{G}'_P(X)) = \left. \frac{d\mathcal{G}'_P(X, P + \theta Y)}{d\theta} \right|_{\theta=0}. \quad (16)$$

Considering (9) and (16), we have $\mathcal{T}_Y(X) = 0$, and

$$\mathcal{T}_Y(A_c^T X A_c) = (\mathcal{T}_Y A_c)^T X A_c + A_c X \mathcal{T}_Y A_c \quad (17)$$

where $A_c \triangleq A - BK$ and $\mathcal{T}_Y A_c = -B \mathcal{T}_Y K$ with

$$\begin{aligned} \mathcal{T}_Y K &= -(R + B^T P B)^{-1} B^T Y B K \\ &\quad + (R + B^T P B)^{-1} B^T Y A \\ &= (R + B^T P B)^{-1} B^T Y A_c. \end{aligned} \quad (18)$$

Substituting (17) and (18) into (16) yields $\mathcal{T}_Y(X - A_c^T X A_c) = -A_c^T X \mathcal{U}_P(Y) - \mathcal{U}_P^T(Y) Y A_c$, which implies that

$$\mathcal{G}''_P(X, Y) = -A_c^T X \mathcal{U}_P(Y) - \mathcal{U}_P^T(Y) X A_c \quad (19)$$

where $\mathcal{U}_P(Y) = B(R + B^T P B)^{-1} B^T Y A_c$. Let $E^{[k]} \triangleq P^{[k]} - P^*$. Consider the second-order Taylor expansion of the residual operator $\mathcal{G}(P^{[k]})$ about the point P^* . We obtain

$$\mathcal{G}(P^{[k]}) = \mathcal{G}'_{P^*}(E^{[k]}) + \frac{1}{2} \mathcal{G}''_{P^*}(E^{[k]}, E^{[k]}) + \mathcal{O}(\|E^{[k]}\|^3). \quad (20)$$

Let $\Delta^{[k]} \triangleq \mathcal{G}'_{P^{[k]}} - \mathcal{G}'_{P^*}$. Since $\mathcal{G}'_{P^{[k]}}$ is Lipschitz continuous with respect to $P^{[k]}$ in a neighborhood of convergence, there exists a constant $c_1 > 0$ such that $\|\Delta^{[k]}\| \leq c_1 \|E^{[k]}\|$. When the iteration enters the convergence region and $\|E^{[k]}\|$ is sufficiently small, it follows that $\|\Delta^{[k]}\| < \|1/(\mathcal{G}'_{P^{[k]}})^{-1}\|$. Thus, by Neumann series inversion, we obtain

$$(\mathcal{G}'_{P^{[k]}})^{-1} = (\mathcal{G}'_{P^*})^{-1} + \mathcal{O}(\|E^{[k]}\|). \quad (21)$$

According to the Taylor expansion in (20), the residual operator satisfies $\mathcal{G}(P^{[k]}) = \mathcal{G}'_{P^*}(E^{[k]}) + \mathcal{O}(\|E^{[k]}\|^2)$. By applying (8) and (21), we have

$$\tilde{P}^{[k+1]} - P^{[k]} = -(\mathcal{G}'_{P^*})^{-1} \mathcal{G}'_{P^*}(E^{[k]}) + \mathcal{O}(\|E^{[k]}\|^2) \quad (22)$$

which implies that $\tilde{P}^{[k+1]} - P^* = \mathcal{O}(\|E^{[k]}\|^2)$. Thus, step 1 ensures that the approximation error is reduced to second

order. Subsequently, the midpoint-based gain update in step 2 yields

$$M^{[k]} = \frac{1}{2} (\tilde{P}^{[k+1]} + P^{[k]}) = P^* + \frac{1}{2} E^{[k]} + \mathcal{O}(\|E^{[k]}\|^2) \quad (23)$$

which implies that $\|M^{[k]} - P^*\| \leq c_2 \|E^{[k]}\|$ for some constant $c_2 > 0$. Moreover, from (12), we have

$$P^{[k+1]} - P^* = E^{[k]} - (\mathcal{G}'_{M^{[k]}})^{-1} \mathcal{G}(P^{[k]}). \quad (24)$$

From (23), we have $\mathcal{G}'_{M^{[k]}} = \mathcal{G}'_{P^*} + 1/2 \mathcal{G}''_{P^*}(X, E^{[k]}) + \mathcal{O}(\|E^{[k]}\|^2)$. Similar to (21), the inverse derivative at the midpoint admits the expansion

$$\begin{aligned} (\mathcal{G}'_{M^{[k]}})^{-1} &= (\mathcal{G}'_{P^*})^{-1} - \frac{1}{2} (\mathcal{G}'_{P^*})^{-1} \mathcal{G}''_{P^*}(X, E^{[k]}) \\ &\quad \times (\mathcal{G}'_{P^*})^{-1} + \mathcal{O}(\|E^{[k]}\|^2). \end{aligned} \quad (25)$$

Then, substituting (20) and (25) into (24) yields

$$(\mathcal{G}'_{M^{[k]}})^{-1} \mathcal{G}(P^{[k]}) = E^{[k]} + \mathcal{O}(\|E^{[k]}\|^3). \quad (26)$$

From (26), it follows that the midpoint structure cancels the second-order term $\mathcal{O}(\|E^{[k]}\|^2)$. Thus, we obtain

$$\begin{aligned} P^{[k+1]} - P^* &= E^{[k]} - (\mathcal{G}'_{M^{[k]}})^{-1} \mathcal{G}(P^{[k]}) = \mathcal{O}(\|E^{[k]}\|^3) \end{aligned}$$

which implies that $\|P^{[k+1]} - P^*\| \leq C \|P^{[k]} - P^*\|^3$, for some scalar constant $C > 0$. Note that there exists a neighborhood of P^* in which the bound in (15) holds. This completes the proof. ■

Remark 3: Algorithm 1 introduces a midpoint Newton update that preserves the geometric structure of the Riccati operator, which makes the iteration more involved than in the scalar case. This added complexity prevents data-storage-based RL methods [5], [6], [7], [8], [9], [10], [11], [12], [13], [14] from being directly extended, as they require repeatedly reusing all past data at each iteration. To overcome this limitation, the next section develops a data-driven accelerated RL algorithm with a verifiable excitation condition that operates without historical data.

IV. DATA-DRIVEN ALGORITHM DESIGN AND ANALYSIS

In this section, we develop a data-driven accelerated RL algorithm based on a verifiable excitation condition to solve Problems (1) and (2). This algorithm does not rely on the restrictive PE condition to guarantee convergence, but only requires that excitation exists within an initial interval. Moreover, the algorithm design is detailed, followed by analyses of the excitation condition and the convergence properties.

Existing data-storage-based RL algorithms utilize all historical data at each iteration to satisfy the PE condition, which is formalized as follows.

Definition 2: PE [42] A bounded regressor $\varpi(\ell) \in \mathcal{R}^p$ is said to be persistently exciting if there exist an integer $T \geq p$ and a constant $a > 0$ such that the following condition holds for all $\ell \geq 0$:

$$\sum_{\tau=0}^{T-1} \varpi(\ell + \tau) \varpi^T(\ell + \tau) \geq a I_{p \times p}. \quad (27)$$

Since (27) depends on the future behavior of $\varpi(\ell)$, it is difficult to verify online. We aim to develop an easily verifiable online condition to evaluate whether the regressor $\varpi(\ell)$ is sufficiently excited, which is defined as follows.

Definition 3: A bounded regressor $\varpi(\ell) \in \mathcal{R}^p$ is said to be initially exciting if there exist an integer $T \geq p$ and $a > 0$ such that the following condition holds over the time interval $[0, T]$:

$$\sum_{\ell=0}^{T-1} \varpi(\ell) \varpi^T(\ell) \geq a I_{p \times p}. \quad (28)$$

Remark 4: Existing continuous-time PE relaxation results [31], [32], [33], [34] cannot be directly applied to our data-driven design. In our formulation, the Lyapunov updates are embedded within a midpoint Newton scheme, which requires constructing new discrete-time filters and data equations that faithfully reproduce the model-based updates, and establishing that the finite-interval positivity of the filtered information matrix ensures the solvability of all data-driven Newton prediction and correction steps. This requirement is different from existing results [31], [32], [33], [34], which do not address midpoint-embedded Lyapunov updates or their discrete-time realizations.

A. Data-Driven Algorithm Design

From Definition 3, the time parameter T is determined by verifying the proposed excitation condition. We now design equivalent data-driven accelerated RL forms based on Algorithm 1. Using the filter outputs to reformulate the proposed accelerated iterative process, our algorithm has three steps.

Step 1: Find data-driven equations that replace (10)–(14). To derive the data-driven algorithm, we rewrite system (1) as

$$x(\ell + 1) = A^{[k]} x(\ell) + B(u(\ell) + K^{[k]} x(\ell)) \quad (29)$$

where $A^{[k]} \triangleq A - BK^{[k]}$. Based on (29), (10) becomes $x^T(\ell) \tilde{P}^{[k+1]} x(\ell) = x^T(\ell) Q x(\ell) + u^T(\ell) R u(\ell) + x^T(\ell + 1) \tilde{P}^{[k+1]} x(\ell + 1)$, where

$$\begin{aligned} x^T(\ell + 1) \tilde{P}^{[k+1]} x(\ell + 1) &= x^T(\ell) (A - BK^{[k]})^T \tilde{P}^{[k+1]} (A - BK^{[k]}) x(\ell) \\ &\quad + x^T(\ell) (A - BK^{[k]})^T \tilde{P}^{[k+1]} B (u(\ell) + K^{[k]} x(\ell)) \\ &\quad + (u(\ell) + K^{[k]} x(\ell))^T B^T \tilde{P}^{[k+1]} x(\ell + 1) \end{aligned} \quad (30)$$

which implies that

$$\begin{aligned} [x^T(\ell) \otimes x^T(\ell) - x^T(\ell + 1) \otimes x^T(\ell + 1)] \text{vec}(\tilde{P}^{[k+1]}) &+ 2 \left[(u(\ell) + K^{[k]} x(\ell))^T \otimes x^T(\ell) \right] \text{vec}(B^T \tilde{P}^{[k+1]} A) \\ &- \left[(K^{[k]} x(\ell))^T \otimes (K^{[k]} x(\ell))^T \right] \text{vec}(B^T \tilde{P}^{[k+1]} B) \\ &+ [u^T(\ell) \otimes u^T(\ell)] \text{vec}(B^T \tilde{P}^{[k+1]} B) \\ &= x^T(\ell) Q x(\ell) + x^T(\ell) (K^{[k]})^T R K^{[k]} x(\ell). \end{aligned} \quad (31)$$

Solving (31) requires recording system data, i.e., $x^T(\ell) \otimes x^T(\ell)$, which is memory-consuming. To reformulate (31) in a compact form, we design the following filters with zero initial conditions:

$$\Upsilon_f(\ell + 1) = \alpha \Upsilon_f(\ell) + f(\ell) \quad (32)$$

$$\Upsilon_{f+}(\ell + 1) = \alpha \Upsilon_{f+}(\ell) + f(\ell + 1) \quad (33)$$

$$\Upsilon_{xx}(\ell + 1) = \alpha \Upsilon_{xx}(\ell) + x(\ell) \otimes x(\ell) \quad (34)$$

$$\Upsilon_{xu}(\ell + 1) = \alpha \Upsilon_{xu}(\ell) + u(\ell) \otimes x(\ell) \quad (35)$$

$$\Upsilon_{uu}(\ell + 1) = \alpha \Upsilon_{uu}(\ell) + f_u(\ell) \quad (36)$$

where $0 < \alpha < 1$ denotes a tunable parameter, $f(\ell) \triangleq \text{vecv}(x(\ell))$, and $f_u(\ell) = \text{vecv}(u(\ell))$. The filter outputs $\Upsilon_f(\ell) \in \mathcal{R}^{0.5n(n+1)}$, $\Upsilon_{f+}(\ell) \in \mathcal{R}^{0.5n(n+1)}$, $\Upsilon_{xx}(\ell) \in \mathcal{R}^{n^2}$, $\Upsilon_{xu}(\ell) \in \mathcal{R}^{nm}$, and $\Upsilon_{uu}(\ell) \in \mathcal{R}^{0.5m(m+1)}$ are obtained from filters (32)–(36). Then, combining (31) and (32)–(36) yields

$$(\varphi^{[k]}(\ell))^T \Pi^{[k+1]} = \Upsilon_f^T(\ell) \text{vecs} \left(Q_1^{[k]} \right) \quad (37)$$

where $Q_1^{[k]} \triangleq (K^{[k]})^T R K^{[k]} + Q$, $\varphi_1^{[k]}(\ell) \triangleq [\varphi_1^T(\ell), (\varphi_2^{[k]}(\ell))^T, (\varphi_3^{[k]}(\ell))^T]^T$, and $\Pi^{[k+1]} \triangleq [\text{vecs}(\Pi_1^{[k+1]})^T, \text{vec}(\Pi_2^{[k+1]})^T, \text{vecs}(\Pi_3^{[k+1]})^T]^T$, where $\Pi_1^{[k+1]} \triangleq \tilde{P}^{[k+1]}$, $\Pi_2^{[k+1]} \triangleq B^T \tilde{P}^{[k+1]} A$, $\Pi_3^{[k+1]} \triangleq B^T \tilde{P}^{[k+1]} B$, $\varphi_1(\ell) \triangleq \Upsilon_f(\ell) - \Upsilon_{f+}(\ell)$, $\varphi_2^{[k]}(\ell) \triangleq 2((d_1^{[k]})^T \Upsilon_{xx}(\ell) + \Upsilon_{xu}(\ell))$, and $\varphi_3^{[k]}(\ell) \triangleq -(d_2^{[k]})^T \Upsilon_{xx}(\ell) + \Upsilon_{uu}(\ell)$ with $d_1^{[k]} \triangleq I_n \otimes (K^{[k]})^T$ and $d_2^{[k]} \triangleq ((K^{[k]})^T \otimes (K^{[k]})^T) D_m$. Then, if (37) is solved, the midpoint gain $\mathcal{K}(M^{[k]})$ can be constructed as follows:

$$\mathcal{K}(M^{[k]}) = \left(R + \frac{(\Pi_3^{[k+1]} + \Theta_3^{[k]})}{2} \right)^{-1} \frac{(\Pi_2^{[k+1]} + \Theta_2^{[k]})}{2} \quad (38)$$

where $\Theta_2^{[k]}$ and $\Theta_3^{[k]}$ are updated by solving the following equation:

$$(\eta^{[k]}(\ell))^T \Theta^{[k+1]} = \Upsilon_f^T(\ell) \text{vecs} \left(Q_2^{[k]} \right) \quad (39)$$

where

$$\begin{aligned} Q_2^{[k]} &\triangleq (K^{[k]})^T R K^{[k]} + Q - \left(\Theta_2^{[k]} \right)^T (K^{[k]} - \mathcal{K}(M^{[k]})) \\ &\quad - (K^{[k]} - \mathcal{K}(M^{[k]}))^T \Theta_2^{[k]} + (K^{[k]})^T \Theta_3^{[k]} K^{[k]} \\ &\quad - (\mathcal{K}(M^{[k]}))^T \Theta_3^{[k]} \mathcal{K}(M^{[k]}) \\ \eta^{[k]}(\ell) &\triangleq \left[\eta_1^T(\ell), (\eta_2^{[k]}(\ell))^T, (\eta_3^{[k]}(\ell))^T \right]^T \\ \Theta^{[k+1]} &\triangleq \left[\text{vecs} \left(\Theta_1^{[k+1]} \right)^T, \text{vec} \left(\Theta_2^{[k+1]} \right)^T, \text{vecs} \left(\Theta_3^{[k+1]} \right)^T \right]^T \end{aligned}$$

where $\Theta_1^{[k+1]} \triangleq P^{[k+1]}$, $\Theta_2^{[k+1]} \triangleq B^T P^{[k+1]} A$, $\Theta_3^{[k+1]} \triangleq B^T P^{[k+1]} B$, $\eta_1(\ell) \triangleq \Upsilon_f(\ell) - \Upsilon_{f+}(\ell)$, $\eta_2^{[k]}(\ell) \triangleq 2((z_1^{[k]})^T \Upsilon_{xx}(\ell) + \Upsilon_{xu}(\ell))$, and $\eta_3^{[k]}(\ell) \triangleq -(z_2^{[k]})^T \Upsilon_{xx}(\ell) + \Upsilon_{uu}(\ell)$ with $z_1^{[k]} \triangleq I_n \otimes (\mathcal{K}(M^{[k]}))^T$ and $z_2^{[k]} \triangleq ((\mathcal{K}(M^{[k]}))^T \otimes (\mathcal{K}(M^{[k]}))^T) D_m$. Then, once (39) is solved, the control gain $K^{[k+1]}$ is updated as follows:

$$K^{[k+1]} = \left(R + \Theta_3^{[k+1]} \right)^{-1} \Theta_2^{[k+1]}. \quad (40)$$

Note that the model-based update steps (10)–(14) are now replaced by their data-driven counterparts in (37)–(40). We next design a verifiable excitation condition to ensure that (37) and (39) are solvable.

Step 2: Design a verifiable excitation to ensure solvability of (37) and (39). Solving $\Pi^{[k+1]}$ and $\Theta^{[k+1]}$ from the least-squares equations (37) and (39) requires that $\varphi(\ell) \varphi^T(\ell)$ and

$\eta(\ell) \eta^T(\ell)$ remain positive definite. Accordingly, we construct the following filters:

$$\Omega(\ell + 1) = \delta \Omega(\ell) + \varphi(\ell) \varphi^T(\ell) \quad (41)$$

$$\Psi(\ell + 1) = \delta \Psi(\ell) + \varphi(\ell) \Upsilon_f^T(\ell) \quad (42)$$

$$\tilde{\Omega}(\ell + 1) = \delta \tilde{\Omega}(\ell) + \eta(\ell) \eta^T(\ell) \quad (43)$$

$$\tilde{\Psi}(\ell + 1) = \delta \tilde{\Psi}(\ell) + \eta(\ell) \Upsilon_f^T(\ell) \quad (44)$$

where $0 < \delta < 1$ is a tunable parameter. $\Omega(\ell), \tilde{\Omega}(\ell) \in \mathcal{R}^{r \times r}$ and $\Psi(\ell), \tilde{\Psi}(\ell) \in \mathcal{R}^{r \times 0.5n(n+1)}$ denote the outputs of the filters (41)–(44) under zero initial conditions, where $r \triangleq 0.5n(n+1) + nm + 0.5m(m+1)$. Then, we present a verifiable excitation condition in the following lemma to ensure that $\Omega(\ell)$ and $\tilde{\Omega}(\ell)$ remain positive definite, which guarantees the solvability of (37) and (39).

Lemma 3: For the regressor $X(\ell) \triangleq [\Upsilon_f^T(\ell), \Upsilon_{xu}^T(\ell), \Upsilon_{uu}^T(\ell)]^T \in \mathcal{R}^r$, a necessary and sufficient condition for $X(\ell)$ to be initially exciting is that the following condition holds at $\ell = T$:

$$\mathcal{X}(\ell) \triangleq \begin{bmatrix} \Lambda_{ff}(\ell) & \Lambda_{fxu}(\ell) & \Lambda_{fuu}(\ell) \\ * & \Lambda_{xuxu}(\ell) & \Lambda_{xuui}(\ell) \\ * & * & \Lambda_{uuuu}(\ell) \end{bmatrix} > 0 \quad (45)$$

where the matrices $\Lambda_{ff}(\ell) \in \mathcal{R}^{0.5n(n+1) \times 0.5n(n+1)}$, $\Lambda_{fxu}(\ell) \in \mathcal{R}^{0.5n(n+1) \times nm}$, $\Lambda_{fuu}(\ell) \in \mathcal{R}^{0.5n(n+1) \times 0.5m(m+1)}$, $\Lambda_{xuxu}(\ell) \in \mathcal{R}^{nm \times nm}$, $\Lambda_{xuui}(\ell) \in \mathcal{R}^{nm \times 0.5m(m+1)}$, and $\Lambda_{uuuu}(\ell) \in \mathcal{R}^{0.5m(m+1) \times 0.5m(m+1)}$ are generated by the following filters under zero initial conditions:

$$\Lambda_{ff}(\ell + 1) = \delta \Lambda_{ff}(\ell) + \Upsilon_f(\ell) \Upsilon_f^T(\ell) \quad (46)$$

$$\Lambda_{fxu}(\ell + 1) = \delta \Lambda_{fxu}(\ell) + \Upsilon_f(\ell) \Upsilon_{xu}^T(\ell) \quad (47)$$

$$\Lambda_{fuu}(\ell + 1) = \delta \Lambda_{fuu}(\ell) + \Upsilon_f(\ell) \Upsilon_{uu}^T(\ell) \quad (48)$$

$$\Lambda_{xuxu}(\ell + 1) = \delta \Lambda_{xuxu}(\ell) + \Upsilon_{xu}(\ell) \Upsilon_{xu}^T(\ell) \quad (49)$$

$$\Lambda_{xuui}(\ell + 1) = \delta \Lambda_{xuui}(\ell) + \Upsilon_{xu}(\ell) \Upsilon_{uu}^T(\ell) \quad (50)$$

$$\Lambda_{uuuu}(\ell + 1) = \delta \Lambda_{uuuu}(\ell) + \Upsilon_{uu}(\ell) \Upsilon_{uu}^T(\ell). \quad (51)$$

Proof: Note that $\mathcal{X}(\ell) = \sum_{\tau=0}^{\ell-1} \delta^{\ell-\tau-1} X(\tau) X^T(\tau)$, which indicates that $\mathcal{X}(\ell)$ is a filtered version of $X(\ell) X^T(\ell)$. If $\mathcal{X}(\ell)$ remains positive definite at $\ell = T$ with $T \geq r$ and satisfies $\mathcal{X}(T) \geq \vartheta I_{r \times r}$ for some constant $\vartheta > 0$, then the following excitation condition is guaranteed:

$$\mathcal{X}(T) = \sum_{\tau=0}^{T-1} \delta^{T-\tau-1} X(\tau) X^T(\tau) \leq \sum_{\tau=0}^{T-1} X(\tau) X^T(\tau) \quad (52)$$

which implies that $X(\ell)$ is initially exciting at $\ell = T$. Now, suppose that $X(\ell)$ is initially exciting at $\ell = T$. By Definition 3, we have

$$\sum_{\tau=0}^{T-1} X(\tau) X^T(\tau) \geq \vartheta I_{r \times r}. \quad (53)$$

Then, by employing (53), we obtain

$$\sum_{\tau=0}^{T-1} \delta^{T-\tau-1} X(\tau) X^T(\tau) \geq \delta^{T-1} \sum_{\tau=0}^{T-1} X(\tau) X^T(\tau) \quad (54)$$

which implies that $\mathcal{X}(T) \geq \delta^{T-1} \vartheta I_{r \times r} > 0$. Thus, we have $\mathcal{X}(\ell) > 0$ holds at $\ell = T$. This completes the proof. ■

The excitation condition (45) implies that $X(\ell)$ is initially exciting, as it relies only on filter data and does not involve iterative parameters, i.e., $K^{[k]}$. The following results show that satisfying (45) ensures the solvability of (37) and (39).

Lemma 4: If condition (45) holds at $\ell = T$, then $\Omega(T) > 0$ and $\tilde{\Omega}(T) > 0$.

Proof: The proof is established by contradiction. Suppose that there exists a nonzero vector $W \triangleq [W_1^T, W_2^T, W_3^T]^T$, where $W_1 \triangleq \text{vecs}(w_1)$, $W_2 \triangleq \text{vec}(w_2)$, and $W_3 \triangleq \text{vecs}(w_3)$ such that $(\varphi^{[k]}(T))^T W = 0$. Together with (31), it follows that:

$$\Upsilon_f^T(T) \text{vecs}(\phi_1) + \Upsilon_{xu}^T(T) \text{vec}(\phi_2) + \Upsilon_{uu}^T(T) \text{vecs}(\phi_3) = 0 \quad (55)$$

where

$$\begin{aligned} \phi_1 &\triangleq (A^{[k]})^T w_1 A^{[k]} - w_1 + 2(K^{[k]})^T (A^T w_1 B - w_2) \\ &\quad - (K^{[k]})^T (B^T w_1 B - w_3) K^{[k]} \\ \phi_2 &\triangleq 2(A^T w_1 B - w_2), \phi_3 = B^T w_1 B - w_3. \end{aligned}$$

Since $X(\ell)$ is initially exciting over the interval $[0, T]$, we have

$$\mathcal{X}(T) = \sum_{\tau=0}^{T-1} \delta^{T-\tau-1} X(\tau) X^T(\tau) > 0.$$

We rearrange (55) as $X^T(T) \tilde{\phi} = 0$, where $\tilde{\phi} \triangleq [\text{vecs}(\phi_1)^T, \text{vec}(\phi_2)^T, \text{vecs}(\phi_3)^T]^T$. From (45) and (55), we obtain

$$\sum_{\tau=0}^{T-1} X(\tau) X^T(\tau) \tilde{\phi} = 0 \quad (56)$$

which implies that $\tilde{\phi} = 0$. Then, we obtain $w_3 = B^T w_1 B$ and $w_2 = A^T w_1 B$, which establishes that $(A^{[k]})^T w_1 A^{[k]} - w_1 = 0$. Since $A^{[k]}$ is Schur stable, it follows that $W = 0$, which contradicts the assumption that W is nonzero. Therefore, we conclude that $\varphi^{[k]}(T)$ has full column rank, and $\Omega(T) > 0$. Similarly, it can be shown that $\tilde{\Omega}(T) > 0$. This completes the proof. ■

Lemma 5: If $\mathcal{X}(\ell) > 0$ at $\ell = T$, then $\mathcal{X}(\ell)$ remains positive definite for all $\ell > T$.

Proof: According to (45), we have

$$\mathcal{X}(\ell) \geq \delta^{\ell-1} \sum_{\tau=0}^{\ell-1} X(\tau) X^T(\tau) \geq \delta^{\ell-1} \sum_{\tau=0}^{T-1} X(\tau) X^T(\tau).$$

Since $\mathcal{X}(T)$ is positive definite, it follows that $\sum_{\tau=0}^{T-1} X(\tau) X^T(\tau) \geq \theta I_{r \times r}$. Therefore, for all $\ell > T$

$$\mathcal{X}(\ell) \geq \delta^{\ell-1} \theta I_{r \times r} > 0$$

which completes the proof. ■

Remark 5: The excitation condition (45) can be easily verified online since it is constructed from known filter outputs and naturally satisfies $\mathcal{X}(\ell) \geq 0$. Thus, the condition (45) can be monitored until $\det(\mathcal{X}(\ell)) > 0$ is achieved.

Remark 6: The solvability of each of (37) and (39) is determined by the excitation condition (45) because the constructed regressor $X(\ell)$ is independent of the control gains $K^{[k]}$ and $\mathcal{K}(M^{[k]})$. Lemma 4 further demonstrates that condition (45) guarantees the solvability of these equations. This condition can be ensured by injecting exploration noise into the control

input. Once satisfied, it remains valid, as established by Lemma 5.

Step 3: Solve the least-squares equations (37) and (39). Note that (37) and (39) are structurally similar, except that (37) contains midpoint gain $\mathcal{K}(M^{[k]})$ and (39) contains control gain $K^{[k]}$. When (45) holds, (37) and (39) are solved at $\ell = T$ with $\rho = 1$ and $\rho = 2$, respectively,

$$\Gamma_\rho^{[k+1]} = (\Xi_\rho^{[k]}(\ell))^{-1} \Delta_\rho^{[k]}(\ell) \text{vecs}(\mathcal{Q}_\rho^{[k]}) \quad (57)$$

where $\Gamma_1^{[k+1]} \triangleq \Pi^{[k+1]}$, $\Gamma_2^{[k+1]} \triangleq \Theta^{[k+1]}$, and

$$\begin{aligned} \Xi_\rho^{[k]}(\ell) &\triangleq \begin{bmatrix} \Xi_{11}(\ell) & \Xi_{\rho 12}^{[k]}(\ell) & \Xi_{\rho 13}^{[k]}(\ell) \\ * & \Xi_{\rho 22}^{[k]}(\ell) & \Xi_{\rho 23}^{[k]}(\ell) \\ * & * & \Xi_{\rho 33}^{[k]}(\ell) \end{bmatrix} \\ \Delta_\rho^{[k]}(\ell) &\triangleq \left[\Delta_1^T(\ell) \left(\Delta_{\rho 2}^{[k]}(\ell) \right)^T \left(\Delta_{\rho 3}^{[k]}(\ell) \right)^T \right]^T \end{aligned}$$

with

$$\begin{aligned} \Xi_{11}(\ell) &\triangleq \Lambda_{ff}(\ell) - \Lambda_{ff+}(\ell) - \Lambda_{f+f}(\ell) + \Lambda_{f+f+}(\ell) \\ \Xi_{\rho 12}^{[k]}(\ell) &\triangleq 2\Lambda_{fxx}(\ell) \mathcal{L}_{\rho 1}^{[k]} + 2\Lambda_{fxu}(\ell) \\ &\quad - 2\Lambda_{f+xx}(\ell) \mathcal{L}_{\rho 1}^{[k]} - 2\Lambda_{f+xu}(\ell) \\ \Xi_{\rho 13}^{[k]}(\ell) &\triangleq -\Lambda_{fxx}(\ell) \mathcal{L}_{\rho 2}^{[k]} + \Lambda_{fuu}(\ell) \\ &\quad + \Lambda_{f+xx}(\ell) \mathcal{L}_{\rho 2}^{[k]} - \Lambda_{f+uu}(\ell) \\ \Xi_{\rho 22}^{[k]}(\ell) &\triangleq 4(\mathcal{L}_{\rho 1}^{[k]})^T \Lambda_{xxx}(\ell) \mathcal{L}_{\rho 1}^{[k]} + 4(\mathcal{L}_{\rho 1}^{[k]})^T \Lambda_{xxu}(\ell) \\ &\quad + 4\Lambda_{uxx}(\ell) \mathcal{L}_{\rho 1}^{[k]} + 4\Lambda_{xuu}(\ell) \\ \Xi_{\rho 23}^{[k]}(\ell) &\triangleq -2(\mathcal{L}_{\rho 1}^{[k]})^T \Lambda_{xxx}(\ell) \mathcal{L}_{\rho 2}^{[k]} + 2(\mathcal{L}_{\rho 1}^{[k]})^T \Lambda_{xxu}(\ell) \\ &\quad - 2\Lambda_{uxx}(\ell) \mathcal{L}_{\rho 2}^{[k]} + 2\Lambda_{xuu}(\ell) \\ \Xi_{\rho 33}^{[k]}(\ell) &\triangleq (\mathcal{L}_{\rho 2}^{[k]})^T \Lambda_{xxx}(\ell) \mathcal{L}_{\rho 2}^{[k]} - (\mathcal{L}_{\rho 2}^{[k]})^T \Lambda_{xxu}(\ell) \\ &\quad - \Lambda_{uxx}(\ell) \mathcal{L}_{\rho 2}^{[k]} + \Lambda_{uuu}(\ell) \\ \Delta_1(\ell) &\triangleq \Lambda_{ff}(\ell) - \Lambda_{f+f}(\ell) \\ \Delta_{\rho 2}^{[k]}(\ell) &\triangleq 2(\mathcal{L}_{\rho 1}^{[k]})^T \Lambda_{xxf}(\ell) + 2\Lambda_{xuf}(\ell) \\ \Delta_{\rho 3}^{[k]}(\ell) &\triangleq -(\mathcal{L}_{\rho 2}^{[k]})^T \Lambda_{xxf}(\ell) + \Lambda_{uuf}(\ell) \end{aligned}$$

where $\mathcal{L}_{11}^{[k]} \triangleq d_1^{[k]}$, $\mathcal{L}_{12}^{[k]} \triangleq d_2^{[k]}$, $\mathcal{L}_{21}^{[k]} \triangleq z_1^{[k]}$, and $\mathcal{L}_{22}^{[k]} \triangleq z_2^{[k]}$. The filter outputs $\Lambda_{ff}(\ell)$, $\Lambda_{ff+}(\ell)$, and $\Lambda_{f+f+}(\ell) \in \mathcal{R}^{0.5n(n+1) \times 0.5n(n+1)}$, $\Lambda_{fxx}(\ell)$ and $\Lambda_{f+xx}(\ell) \in \mathcal{R}^{0.5n(n+1) \times n^2}$, $\Lambda_{fxu}(\ell)$ and $\Lambda_{f+xu}(\ell) \in \mathcal{R}^{0.5n(n+1) \times nm}$, $\Lambda_{fuu}(\ell)$ and $\Lambda_{f+uu}(\ell) \in \mathcal{R}^{0.5n(n+1) \times 0.5m(m+1)}$ are generated by the following filters under zero initial conditions:

$$\Lambda_{fv_\rho}(\ell + 1) = \delta \Lambda_{fv_\rho}(\ell) + \Upsilon_f(\ell) \Upsilon_{v_\rho}^T(\ell) \quad (58)$$

$$\Lambda_{f+f+}(\ell + 1) = \delta \Lambda_{f+f+}(\ell) + \Upsilon_{f+}(\ell) \Upsilon_{f+}^T(\ell) \quad (59)$$

$$\Lambda_{v_\rho xx}(\ell + 1) = \delta \Lambda_{v_\rho xx}(\ell) + \Upsilon_{v_\rho}(\ell) \Upsilon_{xx}^T(\ell) \quad (60)$$

$$\Lambda_{v_\rho xu}(\ell + 1) = \delta \Lambda_{v_\rho xu}(\ell) + \Upsilon_{v_\rho}(\ell) \Upsilon_{xu}^T(\ell) \quad (61)$$

$$\Lambda_{v_\rho uu}(\ell + 1) = \delta \Lambda_{v_\rho uu}(\ell) + \Upsilon_{v_\rho}(\ell) \Upsilon_{uu}^T(\ell) \quad (62)$$

where $\rho = 1, 2$, $v_1 \triangleq f$, and $v_2 \triangleq f^+$. Then, by combining the signal filters [i.e., (32)–(36)] with the information filters [i.e., (41)–(44)], Algorithm 2 summarizes the above data-based algorithm design procedure.

The following result shows that Algorithm 2 achieves the same convergence as Algorithm 1.

Algorithm 2 Data-Based Accelerated RL Algorithm**Input:** Stabilizing $K^{[0]}$, and a threshold $\epsilon > 0$.**Output:** Optimal control gain K^* .

- 1 Apply $u(\ell) = -K^{[0]}x(\ell) + e(\ell)$ to system (1) until (45) is satisfied, where $e(\ell)$ denotes an exploration noise.
- 2 **Solve (57) with $\rho = 1$ and update $\mathcal{K}(M^{[k]})$ using (38).**
- 3 **Solve (57) with $\rho = 2$ and update $K^{[k+1]}$ using (40).**
- 4 **Stop if $\|P^{[k+1]} - P^{[k]}\| < \epsilon$, or set $k \leftarrow k + 1$ and go to Step 2.**

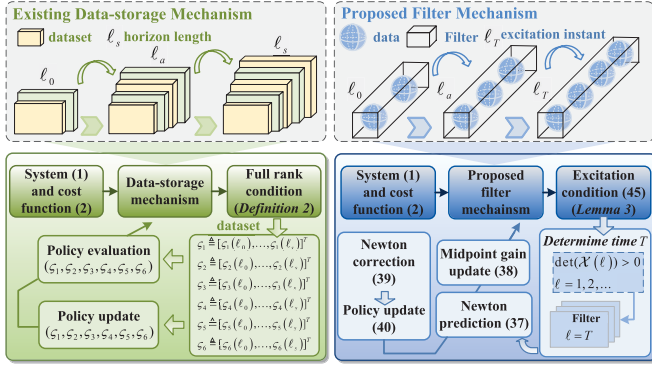


Fig. 3. Comparison between our accelerated RL algorithm and existing data-storage-based RL methods in data processing and iterative computation. The green blocks depict the implementation procedure of existing data-storage-based RL methods, the blue blocks depict the implementation procedure of our accelerated RL algorithm, and the gray blocks highlight the characteristics of their respective data-processing stages.

Theorem 2: With an initial admissible control gain $K^{[0]}$ and condition (45) satisfied, Algorithm 2 implemented via (37)–(40) achieves the same convergence as Algorithm 1.

Proof: We establish the convergence of Algorithm 2 by showing that the data-based computations (37)–(40) are mathematically equivalent to the model-based updates (10)–(14) in Algorithm 1. Since (45) is satisfied, $X(\ell)$ is initially exciting. From Lemma 4, we have $\Omega(\ell) > 0$ and $\tilde{\Omega}(\ell) > 0$, which in turn guarantees that Algorithm 2 yields unique solutions $\Pi^{[k+1]}$ and $\Theta^{[k+1]}$ from (37) and (39). Moreover, (37) and (39) are derived from the model-based updates (10) and (13). By Theorem 1, Algorithm 2 exhibits the same convergence as Algorithm 1. This completes the proof. ■

Remark 7: In Algorithm 2, two types of filters are used. First, the signal filters (32)–(36) generate filtered trajectories that reproduce the model-based midpoint Newton Lyapunov updates along a single trajectory without storing historical data. Second, the information filters [e.g., (41)–(44)] aggregate these filtered trajectories into finite-interval information matrices that satisfy an easily verifiable excitation condition.

B. Discussion

As shown in Fig. 3, unlike data-storage-based RL methods, Algorithm 2 does not require storing historical data. For instance, the output of the designed filter (32) can be written as $\Upsilon_f(\ell) = \sum_{\tau=0}^{\ell-1} \alpha^{\ell-1-\tau} f(\tau)$, which indicates that the filter output $\Upsilon_f(\ell)$ is a weighted accumulation of all past inputs, with

the weights decaying exponentially over time. This stands in contrast to methods that store the raw input $f(\ell)$ in a dataset.

To demonstrate the performance comparison, we implement the updates (10)–(14) using a data-storage-based RL approach. Specifically, for positive integer s , the collected data yield the following matrices:

$$\begin{aligned} \varsigma_1 &\triangleq [\text{vecv}(x(\ell_0)) - \text{vecv}(x(\ell_0 + 1)), \dots, \\ &\quad \dots, \text{vecv}(x(\ell_s)) - \text{vecv}(x(\ell_s + 1))]^T \\ \varsigma_2 &\triangleq [x(\ell_0) \otimes x(\ell_0), x(\ell_1) \otimes x(\ell_1), \dots, x(\ell_s) \otimes x(\ell_s)]^T \\ \varsigma_3 &\triangleq [u(\ell_0) \otimes x(\ell_0), u(\ell_1) \otimes x(\ell_1), \dots, u(\ell_s) \otimes x(\ell_s)]^T \\ \varsigma_4 &\triangleq [\text{vecv}(u(\ell_0)), \text{vecv}(u(\ell_1)), \dots, \text{vecv}(u(\ell_s))]^T \\ \varsigma_5 &\triangleq [\text{vecv}(x(\ell_0)), \text{vecv}(x(\ell_1)), \dots, \text{vecv}(x(\ell_s))]^T \\ \varsigma_6 &\triangleq [\text{vecv}(K^{[k]}x(\ell_0)), \dots, \text{vecv}(K^{[k]}x(\ell_s))]^T \end{aligned}$$

where $\ell_0 < \ell_1 < \dots < \ell_s$. Using collected data and (31), we obtain

$$\bar{\varphi}^{[k]}(\ell) \Theta^{[k+1]} = \varsigma_5 \text{vecs}(\mathcal{Q}_1^{[k]}) \quad (63)$$

where $\bar{\varphi}^{[k]}(\ell) \triangleq [\varsigma_1, 2(\varsigma_2 d_1^{[k]} + \varsigma_3), \varsigma_4 - \varsigma_6]$. The control gain $K^{[k+1]}$ can be constructed as

$$K^{[k+1]} = (R + \Theta_3^{[k+1]})^{-1} \Theta_2^{[k+1]}. \quad (64)$$

Solving (63) requires at least $s \geq \chi$ data samples per iteration. Here, s denotes the number of data samples, and $\chi = 0.5n(n + 1) + nm + 0.5m(m + 1)$ represents the number of unknown parameters. The computational complexity of solving the batch least squares problem (63) for $s \geq \chi$ is on the order of $\mathcal{O}(\chi^2 s)$. In Algorithm 2, once the excitation condition (45) is satisfied at time T , (37) and (39) are solved using only the current filter data at time T . Therefore, the computational complexities of solving (37) and (39) are both on the order of $\mathcal{O}(\chi^2)$, which is lower than solving (63).

Remark 8: Although the proposed accelerated RL algorithm introduces an additional midpoint update step, Algorithm 2 reduces computational complexity in two ways. First, its iterative computation depends only on the trajectory at the current time rather than on the entire dataset, which lowers the per-iteration computational burden. Second, Algorithm 2 achieves a cubic convergence rate, and compared with the quadratic convergence rate of existing methods, it requires fewer iterations and thus further reduces the overall computational complexity.

V. DATA-DRIVEN ALGORITHM FOR INITIAL ADMISSIBLE CONTROL POLICY

To execute the proposed Algorithms 1 and 2, knowledge of an initial admissible control gain $K^{[0]}$ is required. We next present a method for computing a stabilizing gain using initially excited data from Algorithm 2, without requiring additional data collection. Inspired by [23], [36], and [37], a discount factor $\sigma \in [0, 1]$ is introduced into the cost function (2) to eliminate the reliance on an initial admissible control gain by gradually increasing σ from 0 to 1. For a control gain $\tilde{K} \in \mathcal{R}^{m \times n}$, we denote the set of admissible control gains as

$\mathcal{I}_\sigma \triangleq \{\tilde{K} | (\sigma)^{1/2} \lambda(A - B\tilde{K}) < 1\}$. Accordingly, the Lyapunov equation (10) is modified as follows:

$$\mathcal{P}^{[k+1]} = \sigma (A - B\tilde{K}^{[k]})^T \mathcal{P}^{[k+1]} (A - B\tilde{K}^{[k]}) + (\tilde{K}^{[k]})^T \tilde{R}\tilde{K}^{[k]} + \tilde{Q} \quad (65)$$

where $\tilde{Q} = \tilde{Q}^T \in \mathcal{R}^{n \times n} \geq 0$ and $\tilde{R} = \tilde{R}^T \in \mathcal{R}^{m \times m} > 0$. According to Lemma 1, (65) admits a unique solution $\mathcal{P}^{[k+1]} \in \mathcal{R}^{n \times n} > 0$ when $\tilde{K}^{[k]} \in \mathcal{I}_\sigma$. The control gain is updated as

$$\tilde{K}^{[k+1]} = (\tilde{R} + \sigma B^T \mathcal{P}^{[k+1]} B)^{-1} \sigma B^T \mathcal{P}^{[k+1]} A. \quad (66)$$

We initialize $\tilde{K}^{[0]} = 0$ and $\sigma^{[0]} = 0$. The proposed method consists of two steps. *Step 1*: For a given $\tilde{K}^{[k]}$, we employ the bisection method to iteratively update the discount factor $\sigma^{[l]}$ such that (65) admits a unique solution $\mathcal{P}^{[k+1]} > 0$, where l denotes the iteration index of the discount factor update in Step 1. *Step 2*: With the discount factor $\sigma^{[l]}$ fixed, we iteratively solve (65) and update (66) using data. The bisection method adopted in Step 1 is standard and can be found in [43]. Both Steps 1 and 2 involve solving the Lyapunov equation (65). Below, we present its data-driven formulation.

Implementation of Steps 1 and 2: By applying the designed filters (32)–(36) and (65) yields

$$(\tilde{\varphi}^{[k]}(\ell))^T \tilde{\Pi}^{[k+1]} = \Upsilon_f^T(\ell) \text{vecs}(\mathcal{Q}^{[k]}) \quad (67)$$

where $\mathcal{Q}^{[k]} \triangleq (\tilde{K}^{[k]})^T \tilde{R} \tilde{K}^{[k]} + \tilde{Q}$, $\tilde{\varphi}^{[k]}(\ell) \triangleq [\tilde{\varphi}_1^T(\ell), (\tilde{\varphi}_2^{[k]}(\ell))^T, (\tilde{\varphi}_3^{[k]}(\ell))^T]^T$, and $\tilde{\Pi}^{[k+1]} \triangleq [\text{vecs}(\tilde{\Pi}_1^{[k+1]})^T, \text{vecs}(\tilde{\Pi}_2^{[k+1]})^T, \text{vecs}(\tilde{\Pi}_3^{[k+1]})^T]^T$, where $\tilde{\Pi}_1^{[k+1]} \triangleq \mathcal{P}^{[k+1]}$, $\tilde{\Pi}_2^{[k+1]} \triangleq B^T \mathcal{P}^{[k+1]} A$, $\tilde{\Pi}_3^{[k+1]} \triangleq B^T \mathcal{P}^{[k+1]} B$, $\tilde{\varphi}_1(\ell) \triangleq \Upsilon_f(\ell) - \sigma^{[l]} \Upsilon_{f+}(\ell)$, $\tilde{\varphi}_2^{[k]}(\ell) \triangleq 2\sigma^{[l]}((\tilde{d}_1^{[k]})^T \Upsilon_{xx}(\ell) + \Upsilon_{xu}(\ell))$, and $\tilde{\varphi}_3^{[k]}(\ell) \triangleq -\sigma^{[l]}(\tilde{d}_2^{[k]})^T \Upsilon_{xx}(\ell) + \sigma^{[l]} \Upsilon_{uu}(\ell)$ with $\tilde{d}_1^{[k]} \triangleq I_n \otimes (\tilde{K}^{[k]})^T$ and $\tilde{d}_2^{[k]} \triangleq ((\tilde{K}^{[k]})^T \otimes (\tilde{K}^{[k]})^T) D_m$. With the excitation condition (45) satisfied, we solve $\tilde{\Pi}^{[k+1]}$ from

$$\tilde{\Pi}^{[k+1]} = (\tilde{\varphi}^{[k]}(\ell) (\tilde{\varphi}^{[k]}(\ell))^T)^{-1} \tilde{\varphi}^{[k]}(\ell) \Upsilon_f^T(\ell) \text{vecs}(\mathcal{Q}^{[k]}) \quad (68)$$

where

$$\tilde{\varphi}^{[k]}(\ell) (\tilde{\varphi}^{[k]}(\ell))^T \triangleq \begin{bmatrix} \tilde{\Xi}_{11}(\ell) & \tilde{\Xi}_{12}^{[k]}(\ell) & \tilde{\Xi}_{13}^{[k]}(\ell) \\ * & \tilde{\Xi}_{22}^{[k]}(\ell) & \tilde{\Xi}_{23}^{[k]}(\ell) \\ * & * & \tilde{\Xi}_{33}^{[k]}(\ell) \end{bmatrix}$$

$$\tilde{\varphi}^{[k]}(\ell) \Upsilon_f^T(\ell) \triangleq \begin{bmatrix} \tilde{\Delta}_1^T(\ell) & (\tilde{\Delta}_2^{[k]}(\ell))^T & (\tilde{\Delta}_3^{[k]}(\ell))^T \end{bmatrix}^T$$

with

$$\begin{aligned} \tilde{\Xi}_{11}(\ell) &\triangleq \Lambda_{ff}(\ell) - \sigma^{[l]} \Lambda_{ff+}(\ell) - \sigma^{[l]} \Lambda_{f+f}(\ell) \\ &\quad + (\sigma^{[l]})^2 \Lambda_{f+f+}(\ell) \\ \tilde{\Xi}_{12}^{[k]}(\ell) &\triangleq 2\sigma^{[l]} \Lambda_{fxx}(\ell) \tilde{d}_1^{[k]} + 2\sigma^{[l]} \Lambda_{fxu}(\ell) \\ &\quad - 2(\sigma^{[l]})^2 \Lambda_{f+xx}(\ell) \tilde{d}_1^{[k]} - 2(\sigma^{[l]})^2 \Lambda_{f+xu}(\ell) \\ \tilde{\Xi}_{13}^{[k]}(\ell) &\triangleq -\sigma^{[l]} \Lambda_{fxx}(\ell) \tilde{d}_2^{[k]} + \sigma^{[l]} \Lambda_{fuu}(\ell) \\ &\quad + (\sigma^{[l]})^2 \Lambda_{f+xx}(\ell) \tilde{d}_2^{[k]} - (\sigma^{[l]})^2 \Lambda_{f+uu}(\ell) \\ \tilde{\Xi}_{22}^{[k]}(\ell) &\triangleq 4(\sigma^{[l]})^2 (\tilde{d}_1^{[k]})^T \Lambda_{xxx}(\ell) \tilde{d}_1^{[k]} + 4(\sigma^{[l]})^2 (\tilde{d}_1^{[k]})^T \\ &\quad \times \Lambda_{xxu}(\ell) + 4(\sigma^{[l]})^2 \Lambda_{xuu}(\ell) \tilde{d}_1^{[k]} \\ &\quad + 4(\sigma^{[l]})^2 \Lambda_{xuu}(\ell) \\ \tilde{\Xi}_{23}^{[k]}(\ell) &\triangleq -2(\sigma^{[l]})^2 (\tilde{d}_1^{[k]})^T \Lambda_{xxx}(\ell) \tilde{d}_2^{[k]} + 2(\sigma^{[l]})^2 (\tilde{d}_1^{[k]})^T \end{aligned}$$

$$\begin{aligned} &\times \Lambda_{xuu}(\ell) - 2(\sigma^{[l]})^2 \Lambda_{xuu}(\ell) \tilde{d}_2^{[k]} \\ &\quad + 2(\sigma^{[l]})^2 \Lambda_{xuu}(\ell) \\ \tilde{\Xi}_{33}^{[k]}(\ell) &\triangleq (\sigma^{[l]})^2 (\tilde{d}_2^{[k]})^T \Lambda_{xxx}(\ell) \tilde{d}_2^{[k]} - (\sigma^{[l]})^2 (\tilde{d}_2^{[k]})^T \\ &\quad \times \Lambda_{xuu}(\ell) - (\sigma^{[l]})^2 \Lambda_{xuu}(\ell) \tilde{d}_2^{[k]} \\ &\quad + (\sigma^{[l]})^2 \Lambda_{xuu}(\ell) \\ \tilde{\Delta}_1(\ell) &\triangleq \Lambda_{ff}(\ell) - \sigma^{[l]} \Lambda_{f+f}(\ell) \\ \tilde{\Delta}_2^{[k]}(\ell) &\triangleq 2\sigma^{[l]} (\tilde{d}_1^{[k]})^T \Lambda_{xxf}(\ell) + 2\sigma^{[l]} \Lambda_{xuf}(\ell) \\ \tilde{\Delta}_3^{[k]}(\ell) &\triangleq -\sigma^{[l]} (\tilde{d}_2^{[k]})^T \Lambda_{xxf}(\ell) + \sigma^{[l]} \Lambda_{uuf}(\ell). \end{aligned}$$

Then, the control gain is updated as

$$\tilde{K}^{[k+1]} = (R + \sigma^{[l]} \tilde{\Pi}_3^{[k+1]})^{-1} \sigma^{[l]} \tilde{\Pi}_2^{[k+1]}. \quad (69)$$

We next propose a data-driven admissible policy search method in Algorithm 3. Moreover, the following theorem ensures the control gain $\tilde{K}$ obtained from Algorithm 3 is admissible.

Algorithm 3 Data-Based Admissible Gain Computation

Input: $\tilde{K}^{[0]} = 0$, $\sigma^+ = \sigma^* = 1$, $\sigma^{[0]} = \sigma^- = 0$, and a threshold $\epsilon > 0$.

Output: Admissible gain $\tilde{K}$.

```

1 while  $\sigma^{[l]} - \sigma^* > \epsilon$  do
2   while  $\sigma^{[l]} - \sigma^+ > \epsilon$  do
3     Compute  $\sigma^{[l]} \leftarrow \frac{1}{2}(\sigma^- + \sigma^+)$ .
4     Compute  $\tilde{\Pi}_1$  using (68).
5     if  $\tilde{\Pi}_1 > 0$  then  $\sigma^- \leftarrow \sigma^{[l]}$  else  $\sigma^+ \leftarrow \sigma^{[l]}$ 
6      $l \leftarrow l + 1$ ;
7   end
8    $\sigma^{[l]} \leftarrow \sigma^-$  and  $\sigma^+ \leftarrow 1$ 
9   while  $\|\tilde{K}^{[k+1]} - \tilde{K}^{[k]}\| > \epsilon$  do
10    Compute (68) and update  $\tilde{K}^{[k+1]}$  using (69).
11     $k \leftarrow k + 1$ ;
12  end

```

Theorem 3: Under condition (45), the control gain obtained by Algorithm 3 is admissible.

Proof: The bisection method in Algorithm 3 is applied to determine the maximum discount factor $\sigma^{[l]}$ such that $\tilde{K}^{[k]} \in \mathcal{I}_\sigma$. According to [37, Theorem 2], there exists an integer $\tilde{q}$ such that $\sigma^{[l]} = 1$ for all $l \geq \tilde{q}$. In this case, the iterations of (68) and (69) reduce to

$$(\tilde{\varphi}^{[k]}(\ell))^T \tilde{\Pi}^{[k+1]} = \Upsilon_f^T(\ell) \text{vecs}(\mathcal{Q}^{[k]}) \quad (70)$$

and

$$\tilde{K}^{[k+1]} = (R + \tilde{\Pi}_3^{[k+1]})^{-1} \tilde{\Pi}_2^{[k+1]} \quad (71)$$

where $\tilde{\varphi}^{[k]}(\ell) \triangleq [\tilde{\varphi}_1^T(\ell), (\tilde{\varphi}_2^{[k]}(\ell))^T, (\tilde{\varphi}_3^{[k]}(\ell))^T]^T$, where $\tilde{\varphi}_1(\ell) \triangleq \Upsilon_f(\ell) - \Upsilon_{f+}(\ell)$, $\tilde{\varphi}_2^{[k]}(\ell) \triangleq 2((\tilde{d}_1^{[k]})^T \Upsilon_{xx}(\ell) + \Upsilon_{xu}(\ell))$, and $\tilde{\varphi}_3^{[k]}(\ell) \triangleq -(\tilde{d}_2^{[k]})^T \Upsilon_{xx}(\ell) + \Upsilon_{uu}(\ell)$ with $\tilde{d}_1^{[k]} \triangleq I_n \otimes (\tilde{K}^{[k]})^T$ and $\tilde{d}_2^{[k]} \triangleq ((\tilde{K}^{[k]})^T \otimes (\tilde{K}^{[k]})^T) D_m$. Under condition (45), Theorem 2 guarantees the equivalence between implementing (70) and executing (5). Combined with Lemma 1, $\mathcal{P}^{[k+1]}$ converges to the solution of the Riccati equation

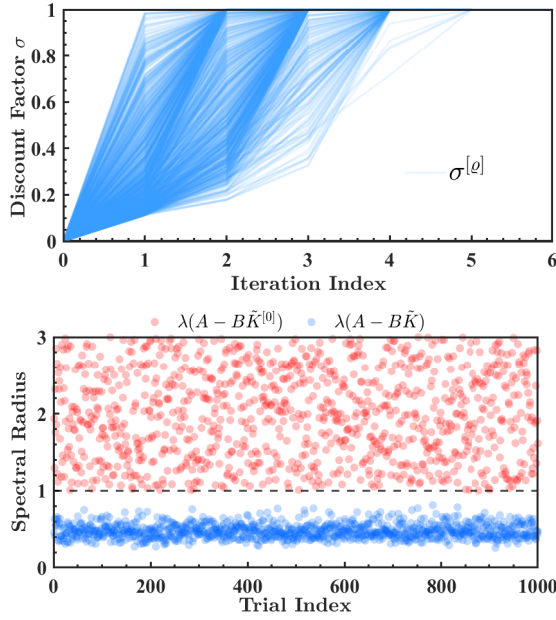


Fig. 4. Monte Carlo simulation of discount factor convergence and spectral radius comparison between open-loop and closed-loop systems in Algorithm 3. Red dots indicate open-loop values. Blue dots indicate closed-loop values.

$A^T \mathcal{P} A - \mathcal{P} - A^T \mathcal{P} B (\tilde{R} + B^T \mathcal{P} B)^{-1} B^T \mathcal{P} A = -\tilde{Q}$. This implies $(A - B\tilde{K})^T \mathcal{P} (A - B\tilde{K}) - \mathcal{P} < 0$, which shows that $\tilde{K}$ is admissible. This completes the proof. ■

Remark 9: It is worth noting that Algorithm 3 requires fewer tuning parameters than existing methods for designing admissible control policies, as it only involves a single parameter σ . Moreover, since the discount factor is embedded in the iterative update (68), the iteration does not require additional system data.

Remark 10: Although a large discount factor may cause the discounted Riccati equation to lose its stabilizing solution, this issue does not occur in Algorithm 3. The discount factor is updated through a bisection procedure together with a feasibility check that enforces the condition $\mathcal{P}^{[k+1]} > 0$. This guarantees that Algorithm 3 identifies the largest discount factor that remains admissible under the stability condition.

VI. SIMULATION

This section demonstrates the effectiveness of Algorithms 1–3 through Monte Carlo experiments and comparative simulations with existing methods.

A. Monte Carlo Simulation

We first employ a Monte Carlo approach to evaluate Algorithm 3. Specifically, we generate 1000 random controllable systems with $n = 10$ and $m = 5$. Each matrix A is sampled from a Gaussian distribution and then scaled such that its spectral radius is uniformly distributed over the interval $[1, 3]$. The matrix B is also Gaussian with full column rank. The matrix $\tilde{Q}$ is constructed as a randomly rotated positive definite diagonal matrix, while $\tilde{R}$ is fixed as I_m . We run Algorithm 3 with the initial settings $\tilde{K}^{[0]} = 0$, $\sigma^{[0]} = 0$,

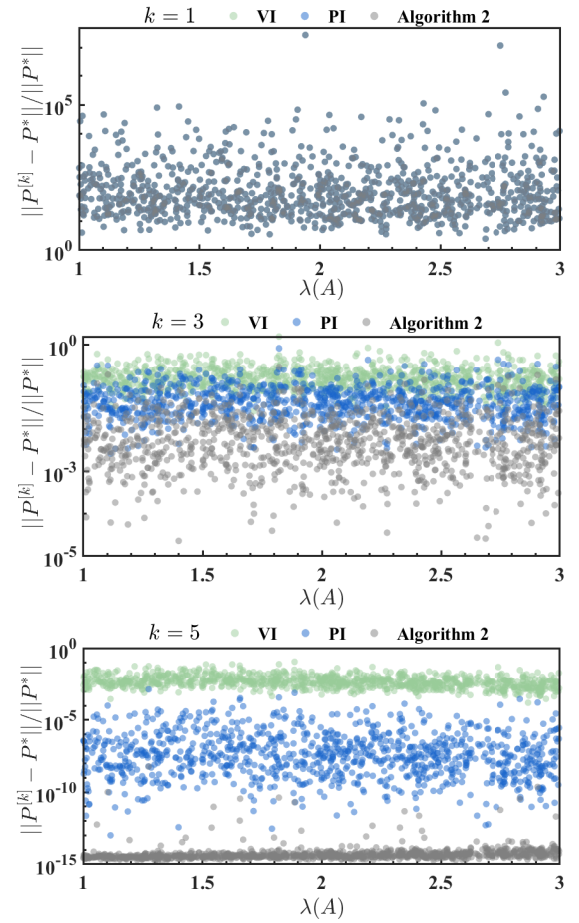


Fig. 5. Monte Carlo relative value error comparisons for VI [41], PI [42], and Algorithm 2 under different $\lambda(A)$ values at iterations $k = 1$, $k = 3$, and $k = 5$. The gray, green, and blue dots represent the relative value errors at the current iteration steps for Algorithm 2, VI [41], and PI [42], respectively.

$\alpha = 0.9$, and $\delta = 0.9$. Fig. 4 illustrates the convergence of $\sigma^{[l]}$, validating the theoretical result in Theorem 3 that $\sigma^{[\bar{l}]} = 1$. Moreover, Fig. 4 shows that the initial closed-loop matrix satisfies $\lambda(A - B\tilde{K}^{[0]}) > 1$, while all converged control gains $\tilde{K}$ lie within the admissible set $\mathcal{I}_\sigma$, for which $\lambda(A - B\tilde{K}) < 1$ holds.

We employ a Monte Carlo method to further evaluate the performance of Algorithm 2, where Algorithm 1 supplies the reference optimal gain P^* for all simulation cases. In each of the 1000 Monte Carlo runs, the system matrix A is generated with a randomly selected spectral radius in $[1, 3]$, B is chosen as a random full-rank matrix, Q is constructed as a random positive-definite matrix, and R is set to $0.5I_m$. To evaluate the convergence performance, Algorithm 2 is compared with VI [41, Algorithm 4] and PI [42, Algorithm 2]. The relative value error is defined as $\|P^{[k]} - P^*\| / \|P^*\|$. As shown in Fig. 5, after five iterations, Algorithm 2 is already much closer to the optimal value, yielding a relative value error on the order of 10^{-15} , in contrast to approximately 10^{-7} for PI and 10^{-3} for VI. Moreover, Fig. 5 indicates that Algorithm 2 requires fewer iterations under the same termination condition and attains higher accuracy when the number of iterations is fixed.

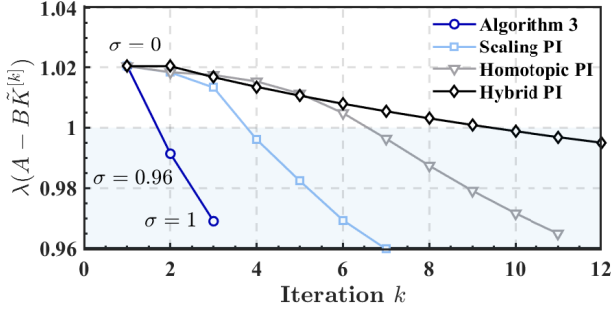


Fig. 6. Evolution of the closed-loop spectral radius $\lambda(A - B\tilde{K}^{[k]})$ under different algorithms.

B. Comparative Simulation

This example compares Algorithms 1–3 with existing methods using a power system model adopted from [44]

$$\begin{bmatrix} \Delta \zeta \\ \Delta \dot{P}_m \\ \Delta f_G \end{bmatrix} = \begin{bmatrix} -\frac{1}{T_g} & 0 & \frac{1}{R_g T_g} \\ \frac{K_t}{T_g} & -\frac{1}{T_t} & 0 \\ 0 & \frac{K_p}{T_p} & -\frac{1}{T_p} \end{bmatrix} \begin{bmatrix} \Delta \zeta \\ \Delta P_m \\ \Delta f_G \end{bmatrix} + \begin{bmatrix} \frac{1}{T_g} \\ 0 \\ 0 \end{bmatrix} \Delta P_c$$

where $\Delta \zeta$, ΔP_m , Δf_G , and ΔP_c represent the incremental changes in governor valve position, generator mechanical power output, frequency deviation, and control input, respectively. The system parameters are given as follows: governor time constant $T_g = 0.08$ s, turbine time constant $T_t = 0.1$ s, generator time constant $T_p = 20$ s, regulation constant $R_g = 2.5$ Hz/MW, generator gain $K_p = 120$ Hz/MW, and turbine gain constant $K_t = 1$ s. Applying the zero-order hold discretization with a sampling interval of 0.01 s yields the following discrete-time model:

$$A = \begin{bmatrix} 0.8838 & 0 & 0.0470 \\ 0.0894 & 0.9048 & 0.0023 \\ 0.0564 & 0 & 1.0009 \end{bmatrix}, B = \begin{bmatrix} 0.1175 \\ 0.0055 \\ 0.0035 \end{bmatrix}$$

which yields $\lambda(A) = 1.0203$. The weighting matrices are selected as $R = 1$ and $Q = I_3$, which yields the optimal control gain $K^* = [0.8506 \ 0.1312 \ 1.3986]$. Three comparative studies are then conducted to highlight the performance of Algorithms 1–3, focusing on admissible policy search (Case I), convergence rate (Case II), and computational complexity (Case III).

Case I: Algorithm 3 is compared with three recently proposed methods that also aim to obtain an initial admissible control gain, namely hybrid PI [15], scaling PI [19], and homotopic PI [21]. For homotopic PI, the parameters are selected as $\tilde{\beta} = 0.1$, $\alpha_0 = 0.0001$, and $\alpha_{j+1} = 0.5\tilde{\alpha}$. For scaling PI, we set $\delta = 0.1$ and $b = 1$. All algorithms are initialized with $\tilde{K}^{[0]} = [0, 0, 0]$ such that $\lambda(A - B\tilde{K}^{[0]}) = 1.0203$. As illustrated in Fig. 6, each algorithm drives the closed-loop system into the stability region. Algorithm 3 exhibits rapid convergence, owing to its two-loop iterative structure and the use of a discount factor σ that varies from 0 to 1.

Case II: To illustrate the difference in convergence rates, Algorithm 2 is compared with PI [42], which exhibit cubic and quadratic convergence, respectively. Both Algorithm 2 and PI are initialized with the same excitation signal and the same

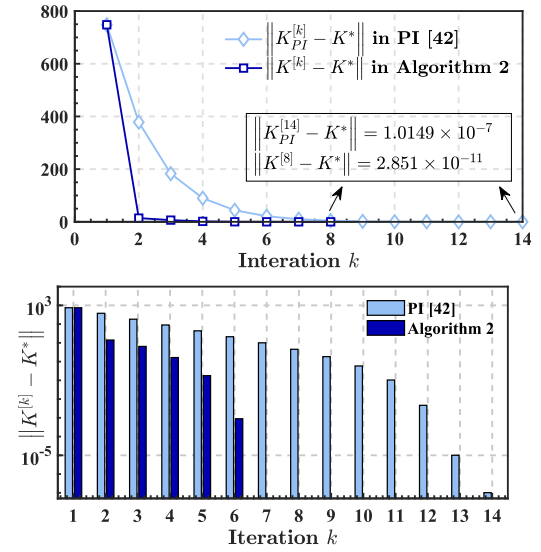


Fig. 7. Comparison of the convergence of $K^{[k]}$ between Algorithm 2 and the PI algorithm [42].

admissible control gain. The parameters are set as $\alpha = 0.9$, $\delta = 0.9$, and $\epsilon = 1 \times 10^{-5}$. Under the same termination accuracy, Fig. 7 shows that Algorithm 2 converges faster and achieves smaller control gain errors compared with the PI method. This improvement results from the additional midpoint computation incorporated in Algorithm 2, which enhances the convergence rate and is consistent with the conclusion of Theorem 2.

Case III: We consider three scenarios to compare the computational complexity of the method in [19] with that of Algorithm 2 under the same initial control gain $\tilde{K} = [0.7054, -0.0001, 1.8771]$ and convergence accuracy $\epsilon = 1 \times 10^{-5}$. Unlike [19], Algorithm 2 includes an additional midpoint computation but does not require storing historical data. In contrast, the method in [19] relies on maintaining past data, and its data-collection duration is typically selected heuristically. Therefore, we evaluate the computational complexity at $\ell = 15$, $\ell = 20$, and $\ell = 25$. As shown in Fig. 8, the execution time of the method in [19] increases steadily as the amount of data grows, indicating higher computational complexity, whereas the runtime of Algorithm 2 remains nearly constant and consistently lower. Although the midpoint update introduces extra computations, it reduces the number of iterations. Together with the fact that Algorithm 2 does not store historical data, its overall computational complexity becomes lower than that of [19]. Moreover, Fig. 9 shows the state trajectories of the power system under Algorithm 2. During the green learning phase, exploration noise is injected into the control input to satisfy condition (45), and at $\ell = 12$, the control gain is updated, after which the system converges steadily.

C. Simulation on Inverted Pendulum System

The dynamics of the inverted pendulum system are described by

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{3(M+m)g}{l(4M+m)} & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ -\frac{3}{l(4M+m)} \end{bmatrix} u(t)$$

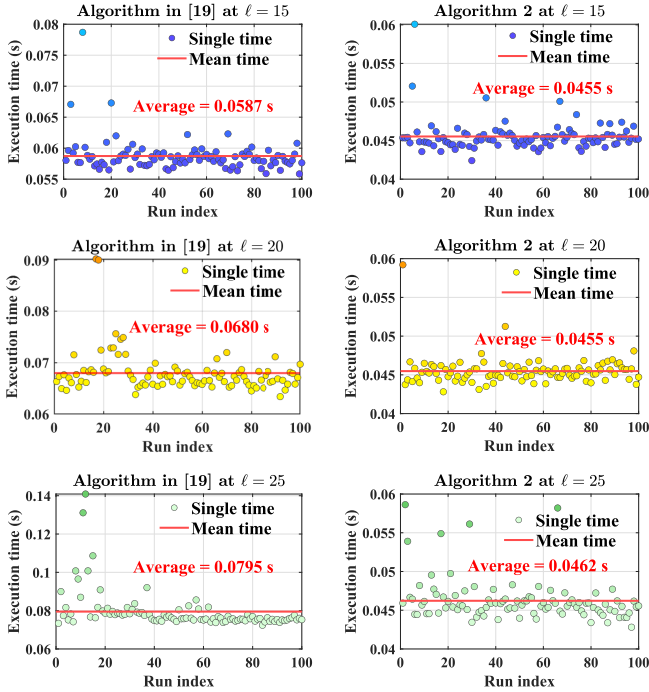


Fig. 8. Computation complexity comparison between Algorithm 2 and the method in [19].

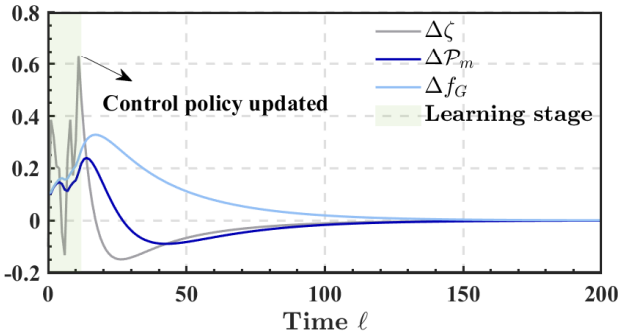


Fig. 9. State trajectories of the power system under $K^{[k]}$ obtained from Algorithm 2. The green-shaded region indicates the learning phase, after which (at $\ell = 12$), the controller is updated to the optimal one.

where $x(t) = [\xi \dot{\xi}]^T$. ξ denotes the deviation of the pendulum from the upright vertical position, while $\mathbb{M}$ and m are the masses of the cart and the pendulum, respectively. The parameter l denotes half of the pendulum length, and g is the gravitational acceleration. Following the settings in [45], we select $\mathbb{M} = 1.096\text{kg}$, $m = 0.109\text{ kg}$, $l = 0.25\text{ m}$, and $g = 9.8\text{ m/s}^2$. With a sampling period $\mathcal{T} = 0.06\text{ s}$, the discretized model is

$$A = \begin{bmatrix} 1.0573 & 0.0611 \\ 1.9284 & 1.0573 \end{bmatrix}, B = \begin{bmatrix} -0.0049 \\ -0.1633 \end{bmatrix}.$$

The weighting matrices are set as $R = 1$ and $Q = I_2$. The optimal control gain is $K^* = [-20.1610 \ -3.6555]$. The exploration noise is selected as $e(\ell) = 0.35 \sin(1.009\ell) + \cos(0.538\ell)^2 + \sin(0.9\ell)$. We first verify the excitation condition (45), and the results are shown in Fig. 10. It can be observed that the matrix $\mathcal{X}(\ell)$ becomes positive definite at

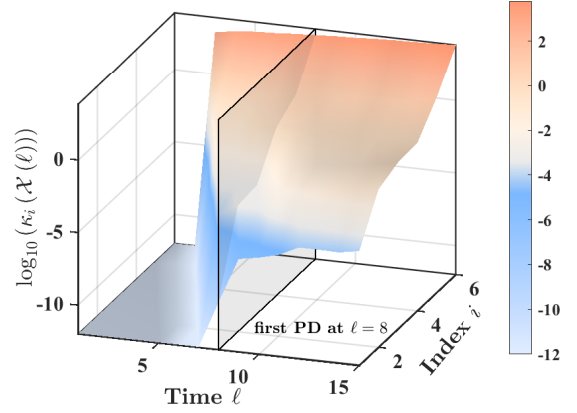


Fig. 10. Eigenvalue evolution of $\mathcal{X}(\ell)$. Each surface point represents $\log_{10}(\kappa_i(\mathcal{X}(\ell)))$ for the i th eigenvalue at time ℓ . The color shading highlights the magnitude variation across eigenvalues and time. The matrix $\mathcal{X}(\ell)$ becomes positive definite for the first time at $\ell = 8$.

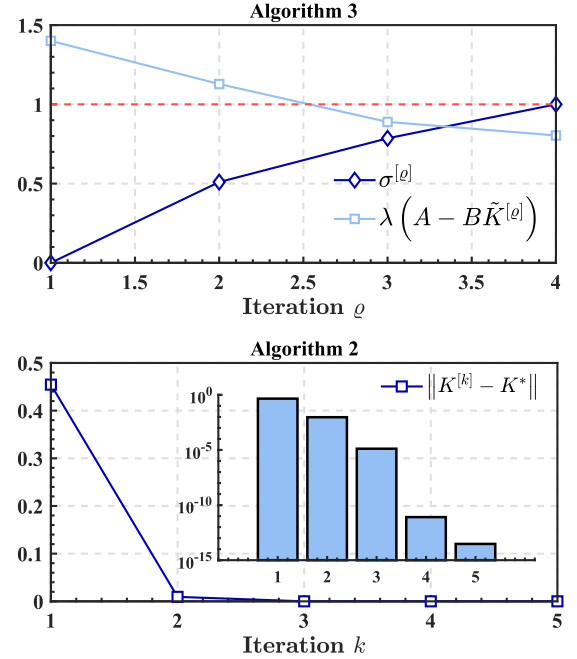


Fig. 11. Convergence of the discount factor $\sigma^{[\ell]}$ and the spectral radius $\lambda(A - B\tilde{K}^{[\ell]})$ in Algorithm 3, and convergence of the control gain $K^{[k]}$ in Algorithm 2.

$\ell = 8\text{ s}$ and remains positive definite afterward, which confirms Lemmas 3 and 5. Fig. 11 further illustrates the procedure in Algorithm 3 for searching an admissible control gain, where the discount factor $\sigma^{[\ell]}$ gradually increases from 0 to 1. In addition, the spectral radius of the closed-loop matrix $\lambda(A - B\tilde{K})$ is less than 1, which verifies Theorem 3. The obtained admissible control gain is $\tilde{K} = [-20.5203 \ -3.9339]$. Fig. 11 shows the convergence of the control gain $K^{[k]}$ to K^* in Algorithm 2 when initialized with $\tilde{K}$, which verifies Theorem 2.

VII. CONCLUSION

In this article, we first developed an accelerated RL algorithm, presented its design details, and rigorously proved

its cubic convergence. Then, we proposed an online verifiable excitation condition to determine whether the system data is sufficiently informative for data-driven control design, and further developed a data-driven method for computing admissible control gains. The convergence of the proposed data-driven algorithm has been rigorously analyzed. Finally, the effectiveness of the proposed method has been validated through randomized Monte Carlo simulations and comparative studies. Our future work will extend the proposed accelerated RL algorithm to multiagent systems and to practical real-world applications.

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